Visual Textualization for Image Prompted Object Detection

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Abstract

We propose VisTex-OVLM, a novel image prompted object detection method that introduces visual textualization —– a process that projects a few visual exemplars into the text feature space to enhance Object-level Vision-Language Models' (OVLMs) capability in detecting rare categories that are difficult to describe textually and nearly absent from their pre-training data, while preserving their pre-trained object-text alignment. Specifically, VisTex-OVLM leverages multi-scale textualizing blocks and a multi-stage fusion strategy to integrate visual information from visual exemplars, generating textualized visual tokens that effectively guide OVLMs alongside text prompts. Unlike previous methods, our method maintains the original architecture of OVLM, maintaining its generalization capabilities while enhancing performance in few-shot settings. VisTex-OVLM demonstrates superior performance across open-set datasets which have minimal overlap with OVLM's pre-training data and achieves state-of-the-art results on few-shot benchmarks PASCAL VOC and MSCOCO. The code will be released at VisTex-OVLM.

1. Introduction

Recently, vision-language models (VLMs) have demonstrated exceptional generalization in CV by pre-training on extensive image-text pairs, allowing zero-shot transfer via natural language prompts and inspiring new approaches for object detection [15, 17, 19, 30, 52, 58]. Among VLMs, object-level VLMs (OVLMs) [5, 31, 33], represented by GLIP [31], achieve better object-text alignment. OVLMs are pre-trained with extensive object detection and phrase grounding data, featuring multi-stage and multi-scale encoders to enhance object-text alignment beyond general VLMs. Furthermore, OVLMs commonly employ a multi-stage encoding structure based on cross-attention, allowing text prompts to actively guide object feature representation and location regression. These characteristics enable OVLMs to surpass conventional VLMs like CLIP [42] in zero-shot transfer for downstream object detection tasks.

However, OVLM's zero-shot object detection (ZSOD) has intrinsic limitations. First, many objects in downstream tasks are underrepresented in the pre-training data, leading to suboptimal zero-shot transfer performance. Second, text prompts often lack sufficient description granularity, resulting in semantic bias and omissions. Third, fine-grained objects share similar descriptions in the text space, making it difficult to distinguish them using text alone [7, 53]. These challenges make introducing new visual information from downstream tasks essential for improving OVLM's transfer ability. One solution is to leverage visual exemplars for fine-tuning, which also refers to few-shot object detection (FSOD) [3, 17, 22, 40, 45]. For example, some VLM-based object detection methods incorporate few-shot visual exemplars information by fine-tuning original weights or adding processing structures. However, these approaches risk disrupting the OVLM's original object-text alignment and harm its generalization capabilities.

To illustrate this, without loss of generality, we conducted an empirical analysis by transferring GLIP, pre-trained on Object365 [44], to MSCOCO [32] using various common transfer methods. Post-transfer, we computed the cosine similarity between paired text and object features from Object365 (Fig. 1). Results revealed that any modification in model structure or weights (methods 2-4) with limited transfer data skewed the object-text similarity distribution in the source domain, partially degrading alignment. This raises the question: can we introduce the semantics of a few visual exemplars while preventing the limited target training data from disrupting OVLM's object-text alignment?

Using images as prompts, i.e., image prompting [36, 53], to provide semantic guidance alongside text prompts may address the question above. However, most OVLMs [5, 31, 33] are text-prompt-exclusive, lacking native support for image prompting due to architectural incompatibilities. MQ-Det [53] is the first to integrate visual exemplars into OVLM's prompting process by adding extra trainable cross-attention

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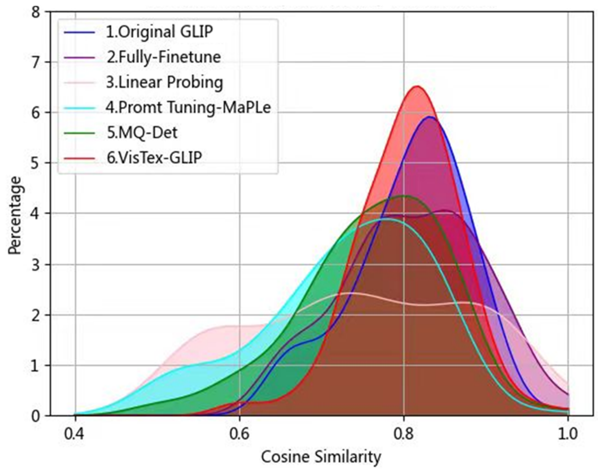

Figure 1. Frequency distribution of feature cosine similarity between object-text pairs from Object365 dataset after transferring the Object365-pretrained GLIP to MSCOCO with different methods.

modules within the text encoder, enabling exemplar images to modulate the text prompts token-wisely and introduce the visual information of novel classes during inference. However, MQ-Det only re-weights tokens of existing text prompts without directly introducing new visual information, which may result in the loss of key clues. Moreover, the newly added cross-attention structures also bias OVLM's pre-trained object-text alignment, as shown in method 5, Fig. 1. How to enable image prompting for text-prompt-exclusive OVLMs without disrupting their pre-trained object-text alignment remains underexplored.

This paper proposes VisTex-OVLM, a novel image prompting method for expanding OVLM's detection capacity beyond pre-trained categories. To preserve OVLM's object-text alignment, we introduce visual textualization, which projects support objects into the text feature space. This process allows limited exemplars from novel categories to participate in prompting, guiding OVLM to perform open-set detection alongside semantic text prompts. The visual textualized support images, combined with the text prompt, are directly input into the unmodified pre-trained OVLM, maintaining its established object-text alignment. Specifically, we first leverage OVLM's visual encoder to extract representations from visual exemplars and design lightweight multi-scale textualizing blocks (MSTBs) for projection. MSTBs process the multi-scale intermediate visual features of the given exemplars produced by each layer of OVLM's visual encoder and uniformly project them into OVLM's text feature space as textualized visual features. Furthermore, we employ a non-parametric multi-stage fusion strategy (MSF) to fuse the textualized visual features from different stages of the visual encoder, leveraging OVLM's multi-stage object-text alignment without introducing additional parameters. Each visual exemplar is projected into a textualized visual token, guiding OVLM in inferring the target image alongside the text prompts. MSTB and MSF fully leverage an OVLM's original structure, robust object-text alignment, and its capability for object-level feature extraction, enabling the textualized visual tokens to achieve optimal semantic representation. During training, MSTB is the only trainable structure. Without fine-tuning on novel classes, MSTB can directly project the visual exemplars from novel categories into semantic-rich textualized visual tokens, which is suitable for practical applications such as open-vocabulary detection on personal mobile devices using image prompts. Previous methods break OVLMs' pre-trained object-text alignment when introducing image prompts. VisTex-OVLM's novelty lies in effectively leveraging the original OVLM structure to incorporate visual information that complements underspecified semantics missing from the text prompts, without altering the model's pre-trained alignment.

Unlike MQ-Det which modulates text features, VisTex-OVLM directly incorporates visual information, preserving OVLM's inference structure and object-text alignment to the greatest extent (method 6, Fig. 1). When applied to GLIP (VisTex-GLIP) and GroundingDINO [33] (VisTex-DINO), VisTex-OVLM demonstrates superior performance in open-set scenarios, including LVIS [16] and 16 datasets with minimal overlap with OVLM pre-training data. Furthermore, VisTex-OVLM also achieves state-of-the-art (SOTA) results on standard few-shot benchmarks, including PASCAL VOC [9] and MSCOCO [32], showing broad applicability. Extensive ablation studies further validate the effectiveness of using images as semantic complements to text prompts. The contributions are summarized as follows:

- We propose VisTex-OVLM, introducing visual textualization, which projects visual exemplars into the text feature space for image prompting to expand OVLM's practical applicability in detecting categories absent from pre-training data without disrupting their object-text alignment.
- Multi-scale textualizing blocks (MSTBs) and a multi-stage fusion (MSF) strategy are designed to preserve the original structure and robust alignment of OVLMs while enabling efficient utilization of their pre-trained knowledge.
- VisTex-OVLM excels with limited visual exemplars in open-set scenarios and achieves SOTA performance on standard FSOD benchmarks.

2. Related works

2.1. Few-shot object detection

Few-shot object detection aims at detecting novel objects utilizing only few-shot annotated instances from novel classes. Mainstream methods mostly rely on complex episode-based meta-learning training paradigms [10, 18, 20, 21, 56] or intricate inter-class relationship modeling [40, 47, 48, 50]. Despite the progress made, FSOD remains a challenging

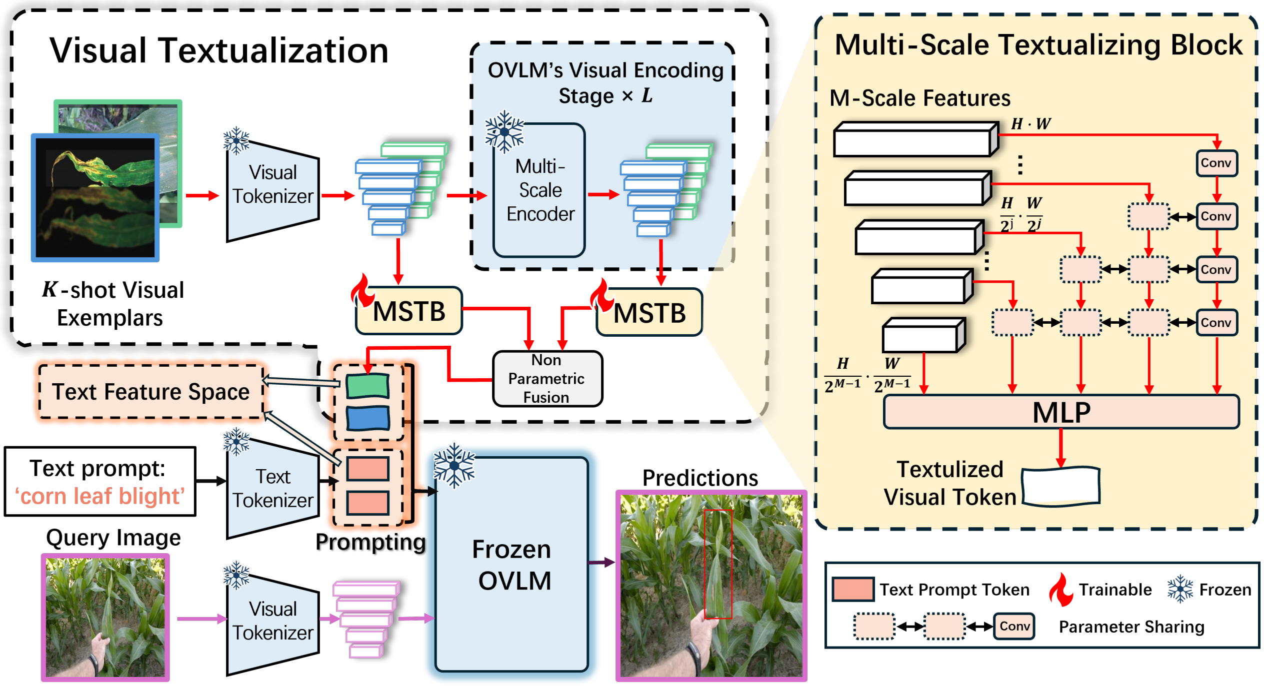

Figure 2. Overview of VisTex-OVLM. VisTex-OVLM performs visual textualization of support images through parameter-shared multi-scale textualizing blocks (MSTB) and a non-parametric multi-stage fusion strategy, mapping support images to text feature space for direct prompting of unmodified OVLMs, such as GLIP [31] and GroundingDINO [33].

task. Recently, VLMs have shown remarkable generalization capabilities, which led to their growing application in FSOD [15, 17, 19, 30, 52, 58]. However, most approaches demonstrate limited performance. The current SOTA model, MTL-FSOD [43], leverages VLMs solely for enhancing classification in FSOD through knowledge distillation, falling short of fully exploiting the potential of VLMs in FSOD. Our VisTex-OVLM probes deeper, aiming to fully leverage the potential of VLM in FSOD.

2.2. VLM-based detection and grounding

VLMs leverage vast text-image pairs obtained from the internet to train its text encoder and image encoder with contrastive learning, allowing the model to learn highly generalizable features. Among them, CLIP [42] is widely recognized for its image-level alignment, supporting effective transfer to image classification and text-image retrieval tasks. However, due to its lack of object-level knowledge, CLIP requires fine-tuning or adding detector modules for detection and grounding tasks [14, 28, 51, 59], increasing model complexity and training costs. Object-level VLMs (OVLMs) designed for detection tasks have emerged to address this [5, 31, 33]. A representative work, GLIP [31] combines object detection and phrase grounding data, utilizing grounded pre-training and a multi-scale, multi-stage cross-attention architecture to establish strong object-text alignment, leading to superior downstream detection and grounding performance over image-level open-vocabulary detectors like regionCLIP [60]. Therefore, our VisTex-OVLM is built upon OVLMs, which are inherently well-suited for detection and grounding tasks, to further explore the potential of VLM on FSOD.

2.3. Prompting methods

Prompting provides task-specific hints to guide large pre-trained models, enhancing their understanding and results [13]. Originating in NLP, it now extends to VLMs.

Design and tuning Prompt design refers to manually constructing prompts [8, 41], which can be complex, time-consuming, and prone to subjective biases. Prompt tuning adds learnable tokens to VLMs and fine-tunes [2, 4, 24, 26, 61, 62]. However, tuning may disrupt VLMs' cross-modal alignment, defecting its generalization capabilities.

Image prompting Image prompting is a novel method where images are regarded as prompts to provide semantic guidance alongside text prompts. CLIPSeg [36] applies this for one-shot segmentation but requires an extra segmentation module, which can disrupt cross-modal alignment. In detection tasks, OWL-ViT [38] allows image prompting, but it relies on image-object pre-training and a predefined architecture and thus cannot be generalized to other OVLMs. MQ-Det [53] integrates a trainable cross-attention module within the text encoder of OVLM, allowing the image prompt to modulate the text prompt in a token-wise manner and thereby introducing the information of prompt images. However, MQ-Det only re-weights tokens without fully incorporating new visual information. Moreover, the additional cross-attention structure may also compromise OVLM's object-text alignment and generalization.

In contrast to the aforementioned approaches, our VisTex-OVLM leverages few-shot support images as prompts through visual textualization, inserting their semantics directly into OVLM's text features. This allows the visual information of image prompts to be directly incorporated into the text prompt. More importantly, our approach does not alter the original structure of OVLM, fully maintaining its excellent object-text alignment, which results in excellent generalization ability and performance.

3. Method

Without loss of generality, we introduce our approach using notations of few-shot object detection (FSOD).

3.1. Task definition

In the classical few-shot object detection (FSOD) setting, there are two distinct datasets: a base set D_b with category set C_b and a novel set D_n with category set C_n, where $C_b \cap C_n = \emptyset$. The base set D_b contains abundant annotated objects that can be utilized for training the FSOD system, while in the novel set D_n, each category $C \in C_n$ has at most K bounding box annotations for K-shot (e.g., $K = 1, 2, 3...10$) object detection learning. In the inference phase, the images providing K-shot annotations are referred to as support images,while the images to be predicted, termed query images, belong to the category set C_n.

In the more practical generalized few-shot object detection (GFSOD) setting, the target category set during inference becomes $C_b \cup C_n$, requiring the model to maintain robust performance on base classes as well.

3.2. Preliminaries and notations for OVLMs

Commonly, OVLMs extend conventional VLMs by incorporating specialized designs for object-level visual tasks [5, 31, 33]. First, they use a hierarchical visual encoder to perform multi-scale encoding and generate region features at different spatial scales—aligned with text features—to capture objects of varying sizes. Second, they expand the text encoder's vocabulary using web-sourced phrase grounding data. Third, they adopt a multi-stage architecture with cross-modal attention to enhance region-text alignment. These specialized designs enable OVLMs to achieve superior object-text alignment, making them particularly suitable for object detection tasks.

To facilitate subsequent discussions, we now define the notation for the OVLM encoding process. Let the intermediate visual features and text features in OVLMs be denoted as R^i and P^i, respectively, where $i = 0, 1, \dots, L$, and L is the total number of stages. R^0 denotes the visual tokens outputted from a pre-trained visual tokenizer, typically a Swin [34], P^0 denotes the text prompt tokens tokenized by a pre-trained text tokenizer, usually BERT [25]. Suppose the text prompt t for target objects contains $|C|$ classes, and for simplicity, assume each class is described using N words ($N = 1$ if only the class name is used as the prompt). Throughout the encoding process, $P \in \mathbb{R}^{|C| \cdot N \times d_T}$, where d_T is the dimension of the text feature space. Due to the multi-scale encoding, $R^i = \left[R^{i,(0)}, ..., R^{i,(j)}, ..., R^{i,(M-1)}\right] = \left[R^{i,(j)}\right]_{j=0}^{M-1}$, where $R^{i,(j)} \in \mathbb{R}^{\frac{H}{2^j} \cdot \frac{W}{2^j} \times d_I}$, H and W are the height and width of the largest scale visual features, M is the total number of scales, and d_I is the visual feature space dimension. The notation $[\cdot]$ represents concatenation.

3.3. VisTex-OVLM

As revealed by the empirical analysis in the introduction, the fine-tuning of weights or additional processing structures tends to compromise an OVLM's well-established object-text alignment, adversely affecting the generalization. To address these challenges, we propose VisTex-OVLM, a novel approach that textualizes support images as prompts during the inference phase to guide a non-finetuned pre-trained OVLM in detecting novel objects, as illustrated in Fig. 2. VisTex-OVLM introduces multi-scale textualizing blocks (MSTB) and a multi-stage fusion strategy to achieve the visual textualization of support images. The MSTB maps the visual features of support images into the text feature space, enabling direct support image prompting of an unmodified OVLM model while preserving its excellent pre-trained object-text alignment.

3.3.1. Visual textualization

An effective approach to modifying VLM's prediction preferences without altering its weights and architecture is to adjust the text prompts in the language branch. Thus, we propose mapping support images to the text feature space, i.e. visual textualization, to incorporate information about novel classes while preserving OVLM's object-text alignment.

Image prompt engineering Since object detection is a dense prediction task, it is crucial to maximize the utilization of few-shot annotations for distinguishing the target object in the support image from irrelevant objects and the background. Following Luddecke et al.'s [36] experience with image prompting in semantic segmentation, we implement image prompt engineering by combining the support images with the target bounding boxes. Specifically, we apply background blur to regions outside the target bounding box.

Multi-scale textualizing Let x_S denote the prompt-engineered one-shot support image. We utilize OVLM's frozen visual encoder to extract multi-scale features containing information about objects of different sizes. For stage i, we denote the extracted visual features of x_S as: $R_S^i = \left[R_S^{i,(j)}\right]_{j=0}^{M-1}$, where $R_S^{i,(j)} \in \mathbb{R}^{\frac{H}{2^j} \cdot \frac{W}{2^j} \times d_I}$. We design a multi-scale textualizing block (MSTB) to process visual features at all scales uniformly and map them to the text feature space, as illustrated in Fig. 2. Specifically, we employ 3×3 convolutions with stride 2 to progressively downsample and extract the most effective visual information from larger-scale features, followed by an MLP that maps the visual features to the text feature dimension d_T.

To reduce the cost of training MSTB on the base set D_b and effectively utilize shared knowledge across features of different scales, we adopt a parameter-sharing strategy. Specifically, we apply a shared set of convolutions to process all features except those at the smallest scale, followed by an MLP mapping to obtain the textualized visual feature $\widetilde{P_S^i}$:

$$\widetilde{R_S^{i,(j)}} = \begin{cases} \left(\prod_j^{M-2} \mathrm{Conv}_{\mathrm{down}}^{i,(j)}\right)\left(R_S^{i,(j)}\right), & \text{if } j \neq M-1 \\ R_S^{i,(j)}, & \text{if } j = M-1 \end{cases} \tag{1}$$

$$\widetilde{R_S^i} = \left[\widetilde{R_S^{i,(j)}}\right]_{j=0}^{M-1} \in \mathbb{R}^{M \cdot \frac{H}{2^{M-1}} \cdot \frac{W}{2^{M-1}} \times d_I}, \tag{2}$$

$$\widetilde{P_S^i} = \mathrm{MLP}^i\left(\widetilde{R_S^i}\right) \in \mathbb{R}^{1 \times d_T}. \tag{3}$$

This strategy facilitates learning the mapping functions that project different-scale features into a unified text feature space. For FSOD, multi-scale textualizing enables comprehensive feature extraction from support samples across different scales, capturing both fine-grained local details and global contextual relationships. This multi-scale approach significantly enhances the representational capability of $\widetilde{P_S^i}$ for target objects in query images.

Multi-stage fusion We further implement a multi-stage fusion (MSF) strategy to integrate textualized visual features $\widetilde{P_S^i}$ from different visual encoder stages of support images. This fusion process consolidates each support image into a single textualized visual token $\widetilde{P_S}$:

$$\widetilde{P_S} = \mathrm{MSF}^i\left(\{\widetilde{R_S^i}\}_{i=0}^{L}\right). \tag{4}$$

This fusion process can be implemented through non-parametric operations, conserving computational resources while achieving effective results. In fact, the synergistic effect of multi-stage fusion and multi-scale textualizing fully leverages OVLM's well-established object-text alignment and object-level visual feature extraction capabilities, enabling the textualized visual token $\widetilde{R_S^i}$ to achieve optimal semantic representation.

3.3.2. Direct support image prompting

Previous methods that use images for prompting either add extra components [36, 53] that risk disrupting OVLM's object-text alignment or modulate text features [53] without directly incorporating visual information. In contrast, VisTex-OVLM achieves direct integration of support image visual information through visual textualization.

In VisTex-OVLM, we concatenate the textualized visual token $\widetilde{P_S}$ with text prompt tokens processed by a pre-trained BERT and input them directly into an unmodified pre-trained OVLM:

$$P^0 = \left[\mathrm{BERT}(t), \widetilde{P_S}\right]. \tag{5}$$

For K-shot tasks, we maintain each shot's independence — essential for OVLM to capture support sample distribution for query prediction — by directly concatenating textualized tokens from different support images, avoiding information-losing fusion:

$$P^0 = \left[\mathrm{BERT}(t), \widetilde{P_{S_1}}, \cdots, \widetilde{P_{S_K}}\right]. \tag{6}$$

For a setting with $|C|$ classes, $P^0 \in \mathbb{R}^{|C| \cdot (N+K) \times d_T}$. During the training phase, we construct corresponding text prompts t using class labels from the base set D_b. Following previous FSOD work, we randomly select K-shot samples as support images for each base class, utilizing the remaining data as queries, and train MSTB end-to-end using OVLM's pre-training loss functions. After training, MSTB can directly perform textualization on novel class data.

Unlike previous image prompting methods, we directly embed the semantics of few-shot support images into OVLM's text prompt P^0 in the form of textualized visual tokens, enabling direct incorporation of visual information. Our approach maintains the OVLM's original architecture, fully preserving its excellent pre-trained object-text alignment. Implementations of VisTex-OVLM on GLIP and GroundingDINO, namely VisTex-GLIP and VisTex-DINO, demonstrate superior generalization ability and performance.

4. Experiment

4.1. Datasets

Open-set scenarios: To rigorously validate VisTex-OVLM's effectiveness on genuinely novel categories, we evaluated its transferability in open-set scenarios, including LVIS MiniVal [16] and 16 unseen datasets with minimal overlap with OVLM pre-training data. LVIS challenges long-tail object detection, while 11 datasets from ODinW35 [29] (mAP≤ 2 for GLIP-L) and 5 medical datasets (MoNu [27], CCRCC [11], ConSep [12], LIDC [1], and Deeplesion [54]) test robustness in non-natural imaging contexts.

Standard FSOD benchmarks Following previous FSOD works [46, 56], we evaluated our method on PASCAL VOC

Method	LVIS MiniVal		Unseen subsets from ODinW35											Unseen medical datasets				
	AP	APr	A	B	C	D	E	F	G	H	I	J	K	MoNu	CCRCC	ConSeP	LIDC	Deeplesion
Meta-DETR [56]	24.6	20.1	22.8	21.5	30.5	23.6	28.9	29.4	23.6	32.4	12.2	23.1	7.3	15.5	18.2	19.9	1.2	0.6
DiGeo [37]	24.1	20.5	35.9	22.6	21.4	25.1	31.5	38.7	29.6	43.1	23.1	21.8	11.6	18.2	20.4	22.8	6.1	2.3
DeFRCN [40]	32.3	28.7	42.3	41.5	31.1	33.8	36.8	41.1	32.6	40.2	33.9	25.2	19.3	16.3	15.6	13.1	5.2	2.8
MFD [50]	28.9	26.4	36.2	33.5	32.3	24.8	27.2	38.5	28.4	34	31.7	42.2	22.8	9.4	13.5	12.1	7.5	3.5
MQ-Det [53]	36.2	31.1	44.8	42.9	38.1	33.5	41.8	40.1	22.6	32.1	22.4	46.3	26.1	7.1	7.6	8.2	6.3	3.1
OWL-ViT [38]	-	-	53.6	42.2	42.7	43.6	40.2	38.4	21.9	38.5	30.5	36.1	23.5	0.2	16.7	15.9	0.2	0.0
OWL-ViTv2 [39]	47.2	37.8	65.4	52.1	45.8	47.8	42.3	50.6	38.2	45.1	38.5	49.8	30.7	4.4	23.4	22.5	6.5	1.2
GroundingDINO-ZS	36.1	30.8	0.1	0.1	0.3	0.3	0.2	0.8	0.4	0.7	1.3	1.3	0.3	2.0	16.2	0.7	0.1	0.0
GroundingDINO-FF*	38.3	33.4	17.5	3.6	6.9	8.2	19.7	7.2	8.1	20.9	4.2	66.9	24.4	25.8	**44.6**	22	2.8	0.5
VisTex-DINO (Ours)	42.8	37.2	70.2	68.4	58.5	66.5	62.0	73.1	56.1	63.3	39.8	71.6	32.5	27.7	38.5	29.6	9.6	10.8
GLIP-ZS	37.3	28.2	0.0	0.0	0.1	0.4	0.9	1.0	0.7	0.1	1.3	0.4	0.4	4.2	8.8	1.3	0.3	0.0
GLIP-FF*	49.5	40.5	75.8	69.8	61.4	**70.5**	67.4	75.4	55.3	63.1	44.5	68.7	**41.2**	28.5	30.6	35.1	11.3	**11.9**
VisTex-GLIP (Ours)	**50.7**	**42.9**	**77.5**	**70.2**	**63.1**	68.0	**69.5**	**75.7**	**58.4**	**65.7**	**46.9**	**73.5**	38.9	**28.9**	31.5	**35.7**	**12.2**	11.7

Table 1. Performance in open-set scenarios. Subsets A~K from ODinW35 [29]: PKLot_640, openPoetryV, boggleBoards, dicemedCol, OxfPetsbybreed, UnoCards, plantdoc, EgoHands_s, webScreenshots, OxfPetsbyspecies, MaskWearing. Best results are marked in **bold**.

Method/Shot		Split 1					Split 2					Split 3					Mean
		1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
non-VLM-based																	
MetaDet [45]	ICCV 19	18.9	20.6	30.2	36.8	49.6	21.8	23.1	27.8	31.7	43.0	20.6	23.9	29.4	43.9	44.1	31.0
MPSR [49]	ECCV 20	41.7	-	51.4	55.2	61.8	24.4	-	39.2	39.9	47.8	35.6	-	42.3	48.0	49.7	44.8
DeFRCN [40]	ICCV 21	53.6	57.5	61.5	64.1	60.8	30.1	38.1	47.0	53.3	47.9	48.4	50.9	52.3	54.9	57.4	51.9
Meta-DETR [56]	TPAMI 22	40.6	51.4	58.0	59.2	63.6	37.0	36.6	43.7	49.1	54.6	41.6	45.9	52.7	58.9	60.6	50.2
MFD [50]	ECCV 22	63.4	66.3	67.7	69.4	68.1	42.1	46.5	53.4	55.3	53.8	56.1	58.3	59.0	62.2	63.7	59.0
ICPE [35]	AAAI 23	54.3	59.5	62.4	65.7	66.2	33.5	40.1	48.7	51.7	52.5	50.9	63.1	55.3	60.6	60.1	55.0
VFA [23]	AAAI 23	57.7	64.6	64.7	67.2	67.4	41.4	46.2	51.1	51.8	51.6	48.9	54.8	56.6	59.0	58.9	56.1
Du et al. [6]	ICCV 23	52.3	55.5	63.1	65.9	66.7	42.7	45.8	48.7	54.8	56.3	47.8	51.8	56.8	60.3	62.4	55.4
SNIDA-MFD [48]	CVPR 24	64.9	67.9	69.7	71.4	70.5	42.2	47.8	54.5	56.6	54.9	58.1	61.3	60.7	63.6	66.0	60.7
VLM-based																	
D&R [30]	AAAI 23	41.0	51.7	55.7	61.8	65.4	30.7	39.0	42.5	46.6	51.7	37.9	47.1	51.7	56.8	59.5	49.3
Norm-VAE [52]	CVPR 23	62.1	64.9	67.8	69.2	67.5	39.9	46.8	54.4	54.2	53.6	58.2	60.3	61.0	64.0	65.5	59.1
FM-FSOD-L [17]	CVPR 24	40.1	53.5	57.0	68.6	72.0	33.1	36.3	48.8	54.8	64.7	39.2	50.2	55.7	63.4	68.1	53.7
DP-DDCL [15]	KBS 24	49.4	60.1	63.9	67.5	69.1	37.5	43.4	48.4	52.4	56.2	45.4	56.3	59.0	63.0	65.7	55.8
VEIC [58]	ESWA 24	67.7	69.5	70.5	70.4	69.1	49.5	51.5	55.0	56.2	54.7	63.3	64.7	63.6	65.4	65.4	62.4
MTL-FSOD [43]	ECCV 24	68.9	71.5	72.1	74.5	72.2	65.5	69.8	**73.5**	**74.4**	73.1	68.8	69.8	70.0	71.6	71.9	71.2
OWL-ViT [38]	ECCV 22	45.3	48.2	51.4	52.3	51.7	42.8	43.1	48.2	51.2	50.3	43.9	45.6	47.3	49.1	49.6	48.0
OWL-ViTv2 [39]	NeurIPS 23	48.6	49.3	51.2	52.6	53.1	45.7	47.8	50.2	52.1	51.4	42.6	46.2	49.6	51.0	51.8	49.5
MQ-Det [53]	NeurIPS 23	53.9	58.4	61.4	64.7	65.1	49.2	51.3	55.6	58.4	60.2	49.9	53.5	57.4	60.2	63.1	57.5
GroundingDINO-ZS	ECCV 24			49.6					43.3					47.4			46.8
GroundingDINO-FF	ECCV 24	49.5	50.4	51.1	52.7	53.1	42.8	43.6	44.8	46.5	47.8	47.5	48.9	50.8	52.4	52.8	49.0
VisTex-DINO	Ours	51.8	53.1	55.8	57	60.2	43.5	46.7	48.5	51.1	53.2	48.6	50.6	53.9	56	58.5	52.6
GLIP-ZS	CVPR 22			50.5					43.0					49.5			47.7
GLIP-FF	CVPR 22	51.3	55.2	59.4	61.2	61.8	44.3	46.2	48.6	49.3	51.2	50.1	52.6	58.3	61.4	61.9	54.2
VisTex-GLIP	Ours	**69.4**	**71.8**	**73.4**	**74.6**	**74.8**	**66.5**	**70.1**	73.1	74.2	**74.4**	**68.9**	**70.2**	**71.1**	**72.3**	**72.6**	**71.8**

Table 2. Comparison of different FSOD methods in terms of AP50 on three PASCAL VOC Novel Split sets. Best results are marked in **bold**.

[9] and MSCOCO [32]. **PASCAL VOC:** We used three partitions of base and novel categories (SPLIT1, SPLIT2, and SPLIT3). Each partition divides the 20 PASCAL VOC categories into 15 base classes (C_b) and 5 novel classes (C_n), reporting AP50 results on the novel set (D_n) under 1, 2, 3, 5, and 10 shots. **MSCOCO:** For MSCOCO's 80 classes, the 20 classes overlapping with PASCAL VOC are designated as novel (C_n), and the remaining 60 as base (C_b), with mAP results reported under 1, 2, 3, 5, 10, and 30 shots.

4.2. Implementation details

Without loss of generality, we conducted main experiments using the pre-trained GLIP-L [31], referred to as VisTex-GLIP. We also implemented VisTex-OVLM on GroundingDINO-T [33], denoted as VisTex-DINO. Unless specified otherwise, we used the following experimental settings for VisTex-GLIP.

The MSTB's MLP consists of two "fully connected (fc) + ReLU" layers. The two fc layers are responsible for spatial aggregation $\left(M \cdot \frac{H}{2^{M-1}} \cdot \frac{W}{2^{M-1}} \rightarrow 1\right)$ and channel dimension transformation ($d_I \rightarrow d_T$), respectively. According to the feature size and scale extracted by the vision encoder of the pre-trained GLIP-L, both H and W are set to 100, and M is 5. By default, MSTB is applied to all 8 stages of GLIP-L, and max pooling across stages is used for non-parametric multi-stage fusion.

For the training of MSTB, we utilized 2 Nvidia RTX 3090 GPUs, each contains 24 GB of memory. For each epoch on

Method		Shot Number					
		1	2	3	5	10	30
non-VLM-based							
Meta R-CNN [55]	ICCV 19	1.0	1.8	2.8	4.0	6.5	11.1
MPSR [49]	ECCV 20	5.1	6.7	7.4	8.7	9.8	14.1
DeFRCN [40]	ICCV21	9.3	12.9	14.8	16.1	18.5	22.6
Meta-DETR [56]	TPAMI 22	7.5	-	13.5	15.4	19.0	22,2
MFD [50]	ECCV 22	10.8	13.9	15.0	16.4	19.4	22.7
Du et al. [6]	ICCV23	-	-	-	-	20.3	22.8
SNIDA-MFD [48]	CVPR 24	12.0	15.4	16.4	17.8	20.7	23.8
VLM-based							
D&R [30]	AAAI23	8.3	12.7	14.3	16.4	18.7	21.8
Norm-VAE [52]	CVPR23	9.5	13.7	14.3	15.9	18.7	22.5
FM-FSOD-L [17]	CVPR 24	5.7	11.0	15.7	21.9	27.7	37.0
DP-DDCL [15]	KBS 24	9.0	12.7	14.6	17.2	19.9	23.0
VEIC [58]	ESWA 24	11.5	14.4	15.3	16.9	19.7	21.9
MTL-FSOD [43]	ECCV 24	12.8	16.9	17.5	19.5	22.7	25.2
OWL-ViT [38]	ECCV 22	22.2	26.1	31.4	32.4	31.5	32.8
OWL-ViTv2 [39]	NeurIPS 23	26.5	28.3	30.5	32.9	33.5	33.1
MQ-Det [53]	NeurIPS 23	42.2	46.3	47.5	47.7	47.8	48.1
GroundingDINO-ZS	ECCV 24	48.5					
GroundingDINO-FF	ECCV 24	48.7	48.9	49.3	50.8	51.6	53.0
VisTex-DINO	Ours	**48.9**	49.2	50.2	52.5	**54.3**	**55.4**
GLIP-ZS	CVPR 22	42.9					
GLIP-FF	CVPR 22	41.2	44.7	47.5	48.8	49.3	49.4
VisTex-GLIP	Ours	47.9	**51.8**	**52.2**	**52.6**	52.7	53.6

Table 3. FSOD performance on MSCOCO. Best results are marked in **bold**.

(D_b), K-shot support images were randomly selected for each class $C \in C_b$, while the rest served as query images. Based on query image labels, the corresponding support images were used to generate textualized image tokens $\widetilde{P_S}$, which were then concatenated with text prompts. The training spanned 30 epochs using AdamW with an initial learning rate of 10^{-4} and weight decay of 0.01.

Detailed implementation for VisTex-DINO and comparison methods are provided in the supplementary materials.

4.3. Open-set scenarios

We evaluated VisTex-OVLM's performance in open-set scenarios, including LVIS MiniVal and 16 unseen datasets with categories exhibiting minimal overlap with the OVLM pre-training data. We compared VisTex-OVLM with methods that have publicly available code and weights, reporting mAP unless otherwise specified in Tab. 1. We established two baselines: OVLM-ZS and OVLM-FF, e.g., GLIP-ZS for GLIP. In OVLM-ZS, we used pre-trained OVLM weights for zero-shot testing. In OVLM-FF, we fine-tuned all pre-trained parameters on random 2-shot support images from the novel dataset, using class names as text prompts for both training and test evaluation, and thus is denoted by the "*" marker in the results. Notably, OVLM-ZS cannot incorporate support images' visual information during testing. Other methods requiring training, including ours, were trained on MSCOCO's base set, treating open-set task sets as novel and testing with 2-shot support images, while OWL-ViT (v2) and OVLM-ZS were evaluated using their frozen pre-trained weights. The results represent the average of five experimental runs. The results show that the zero-shot performance of OVLM significantly drops for novel classes or domains rarely encountered during pre-training compared to its performance on PASCAL VOC and MSCOCO. In contrast, our method outperforms or matches other approaches in most cases, demonstrating strong transferability. It is worth noting that OVLM-FF may achieve slightly higher performance in some cases due to the introduction of new textual knowledge through full fine-tuning. However, VisTex-OVLM, without any training on these datasets, still achieves comparable results. These findings underscore VisTex-OVLM's robust performance in open-set detection and its broad applicability across diverse domains.

Method	10-shot			30-shot		
	AP	bAP	nAP	AP	bAP	nAP
non-VLM-based						
DiGeo [37]	32.0	39.2	10.3	33.1	39.4	14.2
DE-ViT-L [57]	30.6	29.4	34.0	30.6	29.5	34.0
VLM-based						
FM-FSOD-L [17]	40.0	44.2	27.7	43.1	45.2	37.0
MQ-Det [53]	46.9	**46.5**	47.8	47.3	46.8	48.1
VisTex-GLIP	**48.1**	46.2	**52.7**	**48.9**	**47.2**	**53.6**

Table 4. GFSOD performance on MSCOCO. Best results are marked in **bold**.

4.4. Comparison on FSOD benchmarks

We compared our method with other recent FSOD methods, including both non-VLM-based methods and VLM-based methods, with results presented in Tabs. 2 and 3. In OVLM-FF, we fine-tuned all pre-trained parameters on K-shot support images from the novel dataset. Additionally, we directly compared our method with MQ-Det, which is also based on OVLM and uses support images to modulate text prompts for image prompting. Tabs. 2 and 3 demonstrate the superiority of our approach over common transfer methods in utilizing a limited number of support images by textualizing them and directly prompting OVLM, for both GLIP and GroundingDINO.

Our method achieves SOTA performance in classic FSOD while retaining strong recognition for base categories, meeting generalized FSOD (GFSOD) requirements. Tab. 4 shows comparisons with other GFSOD methods on MSCOCO in 10-shot and 30-shot settings. By preserving and leveraging VLMs' generalization, our method avoids overfitting and knowledge forgetting on base classes, achieving high performance across both base and novel categories.

We also conducted compatibility experiments on RegionCLIP and FIBER in the supplementary materials to further validate the generalization ability of VisTex-OVLM.

4.5. Ablation study

We conducted extensive ablation studies on VisTex-OVLM in a 2-shot setting on MSCOCO using GLIP-L, reporting mAP on both the base and the novel classes as in Tab. 4. All other settings were kept at optimal configurations. More

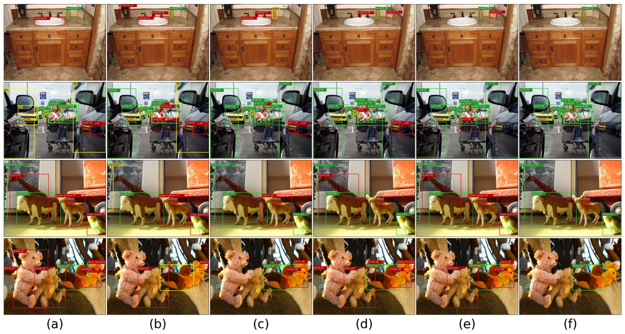

Figure 3. Comparison output visualizations on COCO. (a) Meta-DETR, (b) DiGeo, (c) DeFRCN, (d) MFD, (e) MQ-Det, (f) VisTex-GLIP. Green, red, and yellow boxes denote true positives, false positives, and false negatives.

ablation results and analysis regarding VisTex-OVLM's computational overhead are in the supplementary materials.

We examined the effects of the prompting mode and crucial components, as shown in Tab. 5. Text prompts were generated by concatenating class words following the GLIP paper [31]. We compared GLIP-ZS, GLIP-FF, and GLIP-MaPLe (integrating MaPLe [26]) as a prompt-tuning example. Multi-scale textualization is fundamental to our method, so the "MSTB sharing" column shows only whether parameter-sharing is applied in MSTB. "×" in the "MSF" column for "VisTex-GLIP" indicates that textualization is applied only in stage 1. As detailed in Tab. 5, combining components achieves the best mAP for VisTex-GLIP. Utilizing visual support images addresses GLIP-ZS's limitation of relying solely on text prompts. VisTex-GLIP outperforms GLIP-FF, GLIP-MaPLe, and MQ-Det, which disrupt the object-text alignment by modifying model weights or structure. In contrast, MSTB avoids fine-tuning on novel classes, directly mapping novel support samples into textualized image tokens for prompting, which is user-friendly and avoids disrupting the object-text alignment of GLIP. Notably, image prompts alone without text prompts reduce performance, as text prompts offer essential semantic guidance, impacting VLMs' image-text alignment.

We present attention heatmaps (Fig. 4) showing the effects of different prompting modes and key components, aligned with methods in Tab. 5. These heatmaps visualize object feature attention relative to text features. For "VisTex-GLIP without text prompts", the heatmap represents attention exclusively to textualized visual tokens. The heatmaps show that GLIP-FF and GLIP-MAPLE exhibit significant background noise, while MQ-Det suppresses noise but shows less focused object attention. In contrast, VisTex-GLIP achieves stronger target focus and reduced noise. Notably, VisTex-GLIP without text prompts shows weaker target attention than with text prompts. This observation underscores the importance of text prompts in providing essential semantic guidance, which crucially influences OVLMs' object-text alignment and overall performance.

Method	Text Prompt	Image Prompt	MSTB sharing	MSF	#Par(M)	2-shot AP	2-shot bAP	2-shot nAP
GLIP-ZS	✓	×	×	×	0.00	42.9	44.9	40.5
GLIP-FF	✓	✓*	×	×	397.59	44.3	44.1	44.7
GLIP-MaPLe	✓	✓*	×	×	5.92	48.5	47.8	49.7
MQ-Det	✓	✓	×	×	53.10	44.7	43.5	46.3
VisTex-GLIP	×	✓	×	×	38.52	30.6	32.5	25.9
	✓	✓	×	×	38.52	37.1	29.7	41.2
	✓	✓	✓	×	29.20	48.3	45.7	49.0
	✓	✓	✓	✓	63.06	**50.3**	**48.6**	**51.8**

Table 5. Effectiveness of the prompting mode and crucial components. "✓*" represents directly using support images for fine-tuning, #Par(M) denotes the number of trainable parameters. Best results are marked in **bold**.

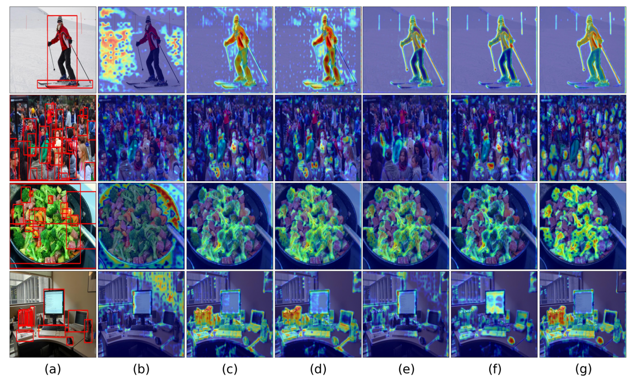

Figure 4. Attention heatmaps comparison. (a) Ground truth, (b) GLIP-ZS (zero-shot), (c) GLIP-FF (fully fine-tuning), (d) GLIP-MaPLe, (e) MQ-Det, (f) VisTex-GLIP without text prompts, (g) VisTex-GLIP.

5. Conclusion

In this paper, we introduce VisTex-OVLM, a novel object detection approach that uses visual textualization to project visual exemplars into the text feature space to prompt an OVLM without disrupting its pre-trained object-text alignment. To implement visual textualization, we design MSTBs and an MSF strategy that fully leverage OVLM's strong object-text alignment and object-level feature extraction capabilities, allowing the textualized visual tokens to achieve optimal semantic representation. VisTex-OVLM achieves state-of-the-art performance on both open-set detection (LVIS and 16 unseen datasets) and few-shot benchmarks (PASCAL VOC and MSCOCO). VisTex-OVLM is applicable to common OVLMs including GLIP and GroundingDINO, underscoring the benefits of using image prompts to complement text prompts and validating its effectiveness in addressing zero-shot object detection limitations.

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